HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2023)

Student No.: 07 Name: _____ Marks: / 46

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. $4^{n+2}3^{2n+4} =$

   A. 6^{2n+4}.

   B. 6^{4n+8}.

   C. 12^{2n+4}.

   D. 12^{3n+6}.

(DSE-2023 PII Q03)

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2. Which of the following statements about the graph of $y = 5 + (x-3)^2$ is true?

A. The graph opens downwards.

B. The x-intercept of the graph is 3.

C. The y-intercept of the graph is 5.

D. The graph passes through the point $(3,5)$.

(DSE-2023 PII Q10)

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3. The scale of a map is $1 : 50000$. If the actual area of an airport is $10\,km^2$, then the area of this airport on the map is

A. $2\,cm^2$.

B. $4\,cm^2$.

C. $20\,cm^2$.

D. $40\,cm^2$.

(DSE-2023 PII Q12)

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4. It is given that z varies as the square of x and the cube root of y. When $x = 12$ and $y = 64$, $z = 36$. When $x = 16$ and $y = 729$, $z =$

A. 108.

B. 144.

C. 162.

D. 216.

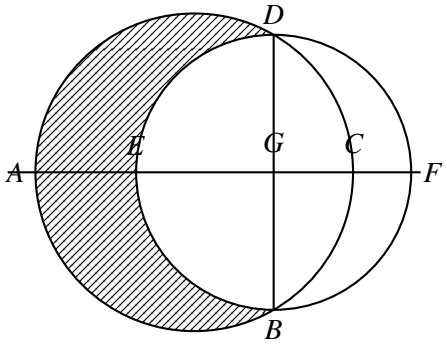
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5. In the figure, AC is a diameter of the circle $ABCD$ while BD and EF are diameters of the circle $BEDF$. It is given that C and E lie on AF. Let G be the point of intersection of AF and BD. If $AG = 30\,\text{cm}$ and $CG = 10\,\text{cm}$, find the area of the shaded region correct to the nearest cm^2.



A. $209\,\text{cm}^2$

B. $367\,\text{cm}^2$

C. $383\,\text{cm}^2$

D. $540\,\text{cm}^2$

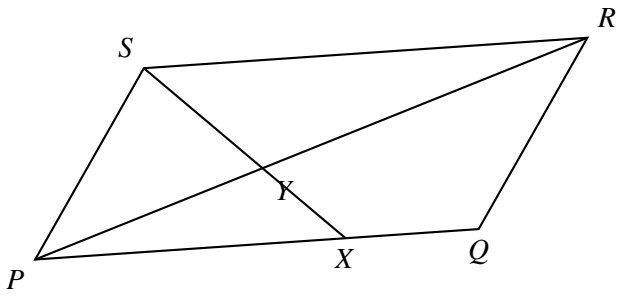
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6. In the figure, $PQRS$ is a parallelogram. Let X be a point lying on PQ. Denote the point of intersection of PR and SX by Y. If the area of $\triangle PXY$ and the area of the quadrilateral $QRYX$ are $32\,\text{cm}^2$ and $58\,\text{cm}^2$ respectively, then the area of $\triangle RSY$ is



A. $40\,\text{cm}^2$.

B. $50\,\text{cm}^2$.

C. $58\,\text{cm}^2$.

D. $72\,\text{cm}^2$.

(DSE-2023 PII Q17)

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7. It is given that $ABCD$ is a rhombus. Denote the point of intersection of AC and BD by E. Which of the following must be true?

I. $AE = BE$

II. $\frac{AE}{AC} = \frac{BE}{BD}$

III. $AE^2 + BE^2 = CD^2$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

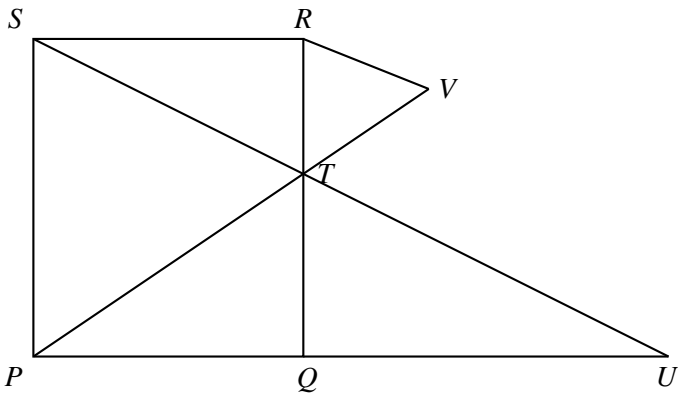
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8. In the figure, $PQRS$ is a rectangle. Let T be a point lying on QR such that $\angle PTS = 90°$. PQ produced and ST produced meet at the point U. PT is produced to the point V such that $RT = RV$. Which of the following must be true?



A. $RV \parallel ST$

B. $\angle PTQ = \angle RTS$

C. $\triangle PST \sim \triangle UTQ$

D. $\triangle PQT \cong \triangle TRS$

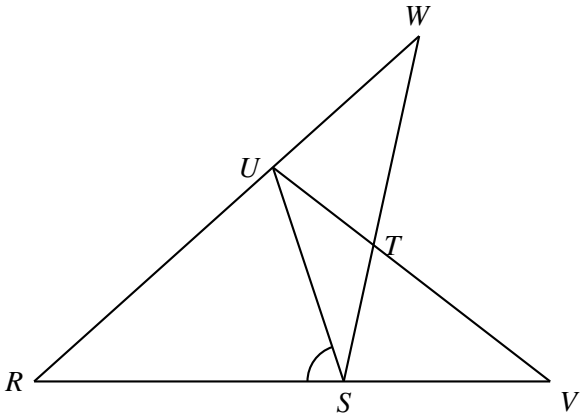
(DSE-2023 PII Q21)

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9. The figure shows the cyclic quadrilateral $RSTU$, where $ST = TU$. RS produced and UT produced meet at the point V while RU produced and ST produced meet at the point W. If $\angle RWS = 32^\circ$ and $\angle RVU = 48^\circ$, then $\angle RSU =$



A. 65°.

B. 73°.

C. 80°.

D. 82°.

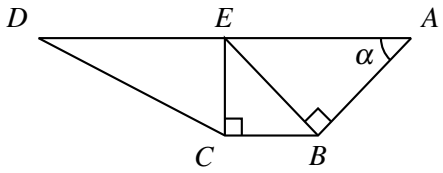
(DSE-2023 PII Q22)

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10. In the figure, $ABCD$ is a trapezium with $AD//BC$. Let E be the mid-point of AD. It is given that $\angle ABE = \angle BCE = 90°$. Find $\dfrac{CE}{DE}$.



A. $\dfrac{1}{2}$

B. 1

C. $\tan\alpha$

D. $\sin\alpha\cos\alpha$

(DSE-2023 PII Q23)

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11. The rectangular coordinates of the point P are $(\sqrt{2}, -\sqrt{2})$. If P is rotated anticlockwise about the origin through $90°$, then the polar coordinates of its image are

A. $(\sqrt{2}, 45°)$.

B. $(\sqrt{2}, 225°)$.

C. $(2, 45°)$.

D. $(2, 225°)$.

(DSE-2023 PII Q24)

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12. Find the constant a such that the straight lines $2x+(a+3)y-5=0$ and $ax-4y+1=0$ are perpendicular to each other.

A. -6

B. -5

C. -2

D. 4

(DSE-2023 PII Q25)

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13. The equations of the straight lines ℓ and L are $9x+12y-37=0$ and $12x+16y+85=0$ respectively. ℓ cuts the x-axis at the point A while L cuts the y-axis at the point B. Let P be a moving point in the rectangular coordinate plane such that the perpendicular distance from P to ℓ is equal to the perpendicular distance from P to L. Denote the locus of P by Γ. Which of the following are true?

I. Γ is parallel to L.

II. Γ is perpendicular to AB.

III. Γ passes through the mid-point of AB.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2023 PII Q26)

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14. The equations of the circles C_1 and C_2 are $x^2+y^2+7x-4y+15=0$ and $2x^2+2y^2-2x-16y-17=0$ respectively. Let G_1 and G_2 be the centres of C_1 and C_2 respectively. Denote the origin by O. Which of the following are true?

I. $\triangle OG_1G_2$ is an equilateral triangle.

II. The line segment OG_1 lies inside C_2.

III. C_1 and C_2 intersect at two distinct points.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2023 PII Q27)

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15. $1011001011001011_2 =$

A. $11 \times 2^{11} + 11 \times 2^5 + 11$.

B. $11 \times 2^{12} + 11 \times 2^6 + 11$.

C. $11 \times 2^{13} + 11 \times 2^7 + 11$.

D. $11 \times 2^{14} + 11 \times 2^8 + 11$.

(DSE-2023 PII Q31)

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16. If k is a real number, then the real part of $\frac{i}{k-i}+\frac{2}{k+i}$ is

A. $\frac{2k+1}{k^2-1}$.

B. $\frac{2k-1}{k^2+1}$.

C. $\frac{k+2}{k^2-1}$.

D. $\frac{k-2}{k^2+1}$.

(DSE-2023 PII Q34)

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17. Let $f(x) = 3x^2 + 18mx + 22m^2$, where m is a real constant. Which of the following statements about the graph of $y = -f(3x)$ must be true?

I. The x-coordinate of the vertex of the graph is m.

II. The y-coordinate of the vertex of the graph is $5m^2$.

III. The equation of the axis of symmetry of the graph is $x + m = 0$.

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2023 PII Q35)

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18. Let $T(n)$ be the nth term of an arithmetic sequence. If $T(11) = 83$ and $T(25) + T(30) = 463$, find the least value of k such that $T(1) + T(2) + T(3) + \cdots + T(k) > 4 \times 10^5$.

A. 299

B. 300

C. 944

D. 945

(DSE-2023 PII Q36)

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19. Consider the following system of inequalities:

$$\begin{cases} x+3 \geq 0 \\ 2x+3y-12 \leq 0 \\ 5x-3y+12 \leq 0 \end{cases}$$

Let D be the region which represents the solution of the above system of inequalities. Find the range of values of β such that the greatest value of $\beta x+6y$ is 24, where (x, y) is a point lying in D.

A. $\beta \leq -10$

B. $\beta \geq -10$

C. $\beta \leq 4$

D. $\beta \geq 4$

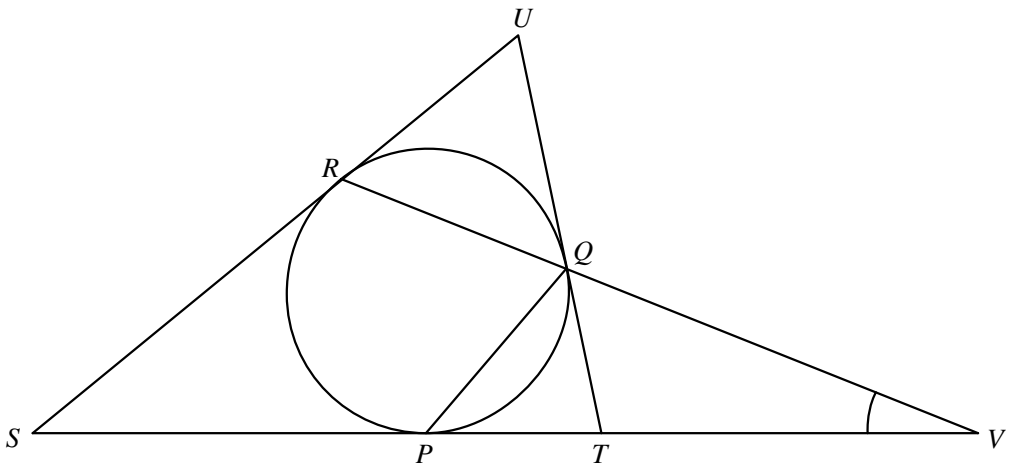
(DSE-2023 PII Q37)

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20. In the figure, P, Q and R are points lying on a circle. ST, TU and SU are the tangents to the circle at P, Q and R respectively. RQ produced and ST produced meet at the point V. If $\angle PSR = 34^\circ$ and $\angle QPT = 46^\circ$, then $\angle PVQ =$



A. 17°.

B. 22°.

C. 27°.

D. 28°.

(DSE-2023 PII Q38)

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21. The equations of the straight lines L_1 and L_2 are $3x-4y+k=0$ and $4x+3y-k=0$ respectively, where k is a positive constant. It is given that L_1 cuts the x-axis at the point P. Denote the point of intersection of L_1 and L_2 by Q. If R is a point lying on L_2 such that the in-centre of $\triangle PQR$ lies on the x-axis, then the x-coordinate of R is

A. $-7k$.

B. $-k$.

C. k.

D. $7k$.

(DSE-2023 PII Q41)

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22. When a boy throws a dart, the probability that he hits the target is 0.6 . If this boy throws the dart 4 times, find the probability that he hits the target at least 2 times.

A. 0.5248

B. 0.7056

C. 0.8208

D. 0.8464

(DSE-2023 PII Q43)

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23. The table below shows the scores (in marks) and the corresponding standard scores of three students in an examination.

Score (marks)	46	x	86
Standard score	-3	1	2

Find x.

A. 64

B. 66

C. 70

D. 78

(DSE-2023 PII Q44)

I can finish _____ questions

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